

Operator Identification for Nonlinear PDEs: MILP and CCP Algorithms

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1 Problem Statement

Given a PDE library $\Theta \in \mathbb{R}^{n \times P}$ (columns are candidate differential operators evaluated on n space-time grid points) and a time-derivative vector $y \in \mathbb{R}^n$, find a sparse coefficient vector $\xi \in \mathbb{R}^P$ such that $y \approx \Theta\xi$. The true PDE has only $k \ll P$ active terms.

1.1 Notation

- n – number of space-time snapshots (e.g. $128 \times 100 = 12800$)
- P – library size (S=12, M=23, L=40, XL=59, ..., P196=196)
- ξ – coefficient vector to recover
- $\|\cdot\|_p$ – standard ℓ_p norm

2 MILP: Mixed-Integer Linear Programming (L0 Baseline)

The MILP approach solves the ℓ_0 -regularized problem via a Big-M formulation and a bisection-based Pareto sweep.

2.1 Formulation

For a fixed tolerance $\varepsilon > 0$, the cardinality minimization is:

$$\min_{\xi, z} \sum_{i=1}^P z_i \quad \text{s.t.} \quad \|y - \Theta\xi\|_1 \leq \varepsilon, \quad |\xi_i| \leq M z_i, \quad z_i \in \{0, 1\} \quad (1)$$

where $M = \max(2\|\xi_{\text{OLS}}\|_\infty, 1)$ is computed from the column-normalized OLS solution.

The ℓ_1 constraint is linearized by introducing slack variables $s \in \mathbb{R}^n$:

$$y - \Theta\xi \leq s, \quad -(y - \Theta\xi) \leq s, \quad \sum_{j=1}^n s_j \leq \varepsilon \quad (2)$$

The resulting MILP has $2P + n$ variables (P continuous, P binary, n continuous slack).

Algorithm 1 Bisection-based Pareto sweep

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1: Input:  $\Theta, y$ , noise floor  $\eta$ , factors  $f_{\text{lo}}, f_{\text{hi}}$ 
2:  $\varepsilon_{\text{lo}} \leftarrow \eta \cdot f_{\text{lo}}$ 
3:  $\varepsilon_{\text{hi}} \leftarrow \eta \cdot f_{\text{hi}}$ 
4:  $k_{\text{lo}}, \xi_{\text{lo}} \leftarrow \text{Solve}(\varepsilon_{\text{lo}})$ 
5:  $k_{\text{hi}}, \xi_{\text{hi}} \leftarrow \text{Solve}(\varepsilon_{\text{hi}})$ 
6:  $K \leftarrow \{k_{\text{lo}} : \varepsilon_{\text{lo}}, k_{\text{hi}} : \varepsilon_{\text{hi}}\}$ 
7: for  $k \leftarrow k_{\text{hi}} + 1$  to  $k_{\text{lo}} - 1$  do
8:    $\varepsilon_L \leftarrow K(k + 1)$   $\triangleright$  tight – not yet feasible at  $k$ 
9:    $\varepsilon_R \leftarrow K(k - 1)$   $\triangleright$  loose – feasible at  $k$ 
10:  for  $t \leftarrow 1$  to  $N_\varepsilon$  do
11:     $\varepsilon_m \leftarrow \sqrt{\varepsilon_L \cdot \varepsilon_R}$ 
12:     $k_{m,-} \leftarrow \text{Solve}(\varepsilon_m)$ 
13:    if  $k_m \leq k$  then  $\varepsilon_R \leftarrow \varepsilon_m$ 
14:    else  $\varepsilon_L \leftarrow \varepsilon_m$ 
15:    end if
16:  end for
17:   $K(k) \leftarrow \varepsilon_R$ 
18: end for
19: Select plateau with smallest  $\varepsilon$  lower bound (tightest tolerance)
20: return  $\xi_{\text{selected}}$ 

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2.2 Bisection-Based Pareto Sweep

Algorithm ?? performs a geometric bisection on ε to trace the Pareto frontier $k(\varepsilon)$ (sparsity vs tolerance).

2.3 Plateau Selection

After the sweep, supports are grouped into runs of consecutive ε values with identical support. The *non-trivial* plateau with the smallest ε lower bound is selected:

$$k^* = \arg \min_{\text{runs}} \varepsilon_{\text{lower}}(r), \quad r \notin \{0\text{-support}\} \quad (3)$$

2.4 Adaptive Tightening

After selecting a support, ε_{lo} is repeatedly halved. If the support remains unchanged for 3 consecutive halvings, the search stops. If the support *changes*, the new support replaces the old one and the counter resets. This catches cases where the true support requires a tighter tolerance than the initial sweep (e.g. Allen–Cahn with $\nu = 0.01$).

3 CCP: Correlation-Cut Pursuit

CCP takes a fundamentally different approach: instead of solving a global MILP, it recursively partitions the feature set by spectral clustering and performs local OLS voting between cluster pairs.

3.1 Algorithm Overview

1. **Correlation graph:** $C_{ij} = |\Theta_i^\top \Theta_j|/n$, diagonal set to zero.
2. **Spectral clustering:** Compute the graph Laplacian $L = D - C$, then its Fiedler vector (second eigenvector of L). Split along the Fiedler ordering recursively until each cluster has $\leq \text{cluster_size}$ features. Stop early if $\lambda_2 > \lambda_{\text{stop}}$ (the subgraph is well-connected).
3. **Cross-group OLS voting:**
For each pair of groups (g_a, g_b) :
 - Concatenate columns into T_{ab}
 - Column-normalize: $T_{ab}/\|T_{ab}\|_2$
 - Solve $\min_{\xi} \|y - T_{ab}\xi\|_2^2$ (OLS)
 - Vote for column j if $|\xi_j| > \tau \cdot \max |\xi|$ ($\tau = 10^{-3}$)
4. **Survivor selection:** Terms with votes $\geq \lceil n_g/2 \rceil$ survive. If nothing pruned, tighten to $n_g - 1$ votes.
5. **Iterate:** Re-cluster survivors; repeat until support $\leq \text{cluster_size}$ or no change.
6. **Final OMP + OLS:** If > 6 survivors, run OMP to pick 6; debias with OLS; prune coefficients below $10^{-4} \cdot \max |coef|$.

3.2 Key Design Decisions

Column normalization. Raw OLS coefficients are dominated by column magnitude (e.g. u^3 has much larger entries than u). Normalizing to unit ℓ_2 norm makes coefficients comparable. This was the key fix that improved CCP from 10% to 95% on Burgers at $P = 40$.

Relative threshold. The threshold $\tau = 10^{-3}$ matches the Lasso pruning heuristic. It filters out terms whose normalized OLS coefficient is below 0.1% of the dominant term. The original threshold 10^{-14} let every non-zero term through, causing false positives at large P .

Fiedler stop. The adaptive stopping criterion $\lambda_2 > \lambda_{\text{stop}}$ prevents over-cutting well-connected subgraphs, avoiding degenerate small clusters that amplify noise.

4 Computational Comparison

MILP (L0 SCIP/L0 CBC):

- $n_{\varepsilon} = 30$ bisection iterations per seed
- ≈ 30 –40 MILP solves per seed
- Each MILP: $2P + m$ variables (P binary), $m = 500$ subsampled rows
- SCIP: 12K LP iterations per solve, 137 cutting planes at root node, ~ 46 s/solve on $P = 40$ (Allen–Cahn)
- Total: 15–30 minutes per seed at $P = 40$

CCP:

- Spectral clustering via LAPACK `dsyevx` with `subset_by_index=[1,1]` (Fiedler vector only)
- $\binom{n_g}{2}$ OLS solves per round ($n_g = P/\text{cluster_size}$)
- Each OLS: $\leq 2 \cdot \text{cluster_size}$ columns $\times$ 12800 rows
- Total: 0.5–10s per seed (scaling roughly $O(P^2)$ from the Fiedler eigenvalue)

5 Benchmark Results

5.1 CCP Across Library Sizes ($P = 12$ to $P = 196$)

Table ?? shows CCP (default `cluster_size=8`) across 16 library sizes, 7 PDEs, 20 seeds each. Values are Jaccard $J = 1$ counts.

Table 1: CCP Jaccard $J = 1$ across library sizes ($n = 20$ seeds per cell)

PDE	12	23	40	59	45	55	75	81	89	96	126	133	146	154	174	196
KdV	20	19	19	20	20	19	19	19	20	20	18	16	19	18	14	16
Burgers	20	20	19	19	20	19	17	16	15	16	13	9	11	15	9	6
AC	–	–	20	20	20	20	19	18	19	18	17	19	18	19	17	17
KS	–	18	16	8	9	8	8	8	5	9	4	4	2	2	3	1
FKPP	20	20	14	9	11	15	12	17	10	10	7	7	3	5	4	3
KdVB	20	20	19	19	19	17	11	10	8	11	7	5	5	4	4	1
FHN	–	–	18	17	19	18	13	9	14	9	10	7	9	4	6	6

KdV and Burgers maintain $J = 1$ up to $P \sim 150$. AC and FHN are near-perfect through $P \sim 170$. KS and FKPP degrade fastest due to high feature correlation. KdVB is fragile above $P \sim 80$ (3 true terms across clusters).

5.2 MILP vs CCP at $P = 40$ (L Library)

Table ?? compares L0 SCIP, L0 CBC, and CCP on the standard L library ($P = 40$), 7 PDEs, 20 seeds.

Table 2: MILP vs CCP at $P = 40$ (Jaccard $J = 1$ / 20 seeds)

PDE	L0 SCIP	L0 CBC	CCP
KdV	20/20	16/20	19/20
Burgers	20/20	20/20	19/20
AC	–	–	20/20
KS	–	–	16/20
FKPP	20/20	19/20	14/20
KdVB	20/20	9/20	19/20
FHN	20/20	–	18/20

Notes:

- SCIP and CBC results from $n_\epsilon = 30$ bisection runs.
- Dashes (–) indicate CoinError solver crash (CBC on AC, KS, FHN) or solver not installed (SCIP on AC, KS).

- KdVB L CBC: 11/20 CoinError failures (pre-marked); SCIP provides $J = 1$ for all seeds.
- SCIP solves take 15–30 min/seed at $P = 40$; CCP takes 0.5–2s/seed.
- CCP outperforms MILP on KS, AC; MILP edges on FKPP, KdVB.

5.3 Solver Comparison: SCIP vs CBC

Both solve the identical Big-M MILP formulation (Eq. ??) with the same subsample ($m = 500$, identical random seed). Differences arise from solver internals:

- **Presolve:** SCIP tightens 1602 variable bounds in 5 rounds; CBC’s presolve is less aggressive.
- **Cutting planes:** SCIP generates 137 cuts in the root node, closing the gap from 621% to 114% before branching. CBC generates fewer, weaker cuts.
- **Heuristics:** SCIP’s “integrality shift” finds feasible solutions faster (primal bound 2.0 found at root). CBC relies more on branching.
- **Branching:** SCIP explores 1–2 nodes (total 2 runs after restart with 28 global fixings). CBC may explore deeper trees.
- **Robustness:** CBC crashes with CoinError on ill-conditioned instances (AC, KS, FHN at $P = 40$). SCIP handles all 7 PDEs without crashing.

5.4 Solver Runtime Breakdown

For a single Allen–Cahn seed ($P = 40$, $n = 12800$):

Phase	SCIP (s)	CBC (s)	CCP (ms)
One MILP solve ($\varepsilon = 100$)	46.3	2–25	–
$\times 30$ bisection steps	1389	60–750	–
Total (one seed)	~ 25 min	~ 2 –12 min	~ 500 ms

CCP’s cross-group OLS voting (~ 28 pairs $\times$ OLS on ≤ 16 columns) replaces the global MILP with ~ 28 cheap LS solves, achieving a $1000\times$ speedup.

6 Discussion

6.1 Why CCP Outperforms MILP on KS/AC

Allen–Cahn ($\nu = 0.01$): The true support has coefficients $\{u^3 : 382, u : 136, u_{xx} : 21\}$. The Big-M $M \approx 764$ treats all three identically. The MILP can drop u_{xx} (coefficient 21) because the ε -tolerance absorbs its residual. More bisection iterations ($n_\varepsilon = 30$) help but don’t fully solve this – the adaptive tightening added in this work addresses it by halving ε_{10} until the support stabilizes.

CCP succeeds because u_{xx} is clustered into a small group (it has low correlation with u and u^3) where OLS easily identifies it.

Kuramoto–Sivashinsky: The library contains many terms correlated with u_{xxxx} ($u^2 u_{xxxx}, u^3 u_{xxxx}, \dots$). The OLS coefficients show no clear separation between true terms ($u_{xxxx} : 3110$) and spurious ($u^2 u_{xxxx} : 479$). The MILP’s Pareto front is flat (many sparsity levels achieve similar ε), making plateau selection unreliable.

CCP's clustering groups correlated terms together; the OLS voting then selects the correct representative from each group.

6.2 When to Use MILP

MILP (SCIP) is useful when:

- The library is small ($P \leq 23$) – solves are fast (2–5s)
- The coefficient magnitudes are well-separated (KdV, Burgers, KdVB)
- A provable optimality certificate is needed
- The CCP voting fails due to wide term spread across clusters

For large libraries ($P \geq 40$), CCP is the preferred method: faster, more robust, and equally or more accurate on most PDEs.